

OMAR TRONELLI

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EDUCATION

Sapienza University of Rome

Rome, Italy

MSc Financial Risk and Data Analysis *GPA: 4.0*

Sep 2024 – Present

Relevant coursework: Quantitative Financial Modeling, Econometrics for Financial Markets,
Probability & Stochastic Processes, Machine Learning

BSc Economics and Finance

Sep 2021 – Oct 2024

Relevant coursework: Econometrics, Advanced Statistics, Monetary Policy,
Financial Intermediaries, EU Institutions and Banking Union

Thesis: *"Predictive Models for Exchange Rates"*

Supervisor: Prof. Marco Di Pietro

ACADEMIC PROJECTS

Financial Modeling and Derivatives

SOFR OIS Yield Curve Construction

- Designed and implemented a full pipeline to construct the SOFR OIS zero-coupon yield curve from par swap rates.
- Developed a numerical bootstrapping framework to recover discount factors via a bisection root-finding algorithm.
- Derived zero rates and constructed a smooth, arbitrage-free term structure using log-linear interpolation.
- Implemented the complete calibration and visualization workflow in Python (rateslib).

Option Pricing and Hedging with Greeks

- Derived the Black–Scholes–Merton option pricing formula using both replication and risk-neutral valuation.
- Analyzed the Greeks and their application in risk management.
- Implemented hedging strategies, focusing on Gamma and Theta, and hedge performance for a short gamma trader.

Volatility Forecasting

- Compared GARCH out-of-sample predictive accuracy, with ML models, for Euro Stoxx 50.
- Focused on the development of an LSTM neural network.
- Evaluated the performance of one-step-ahead predictions.

Applied Statistical Modeling

- Worked on applied statistics projects involving panel-data regression, cross-validation, LASSO model selection, kernel regression, clustering, and multiple regression analysis.

Gaussian Processes

- Collaborating with Dr Francesca Panero and Dr Sahoko Ishida on a project involving Gaussian Process Regression. Responsible for coding; implemented custom kernels in GPflow (TensorFlow-based library), with the goal of performing model selection.

SKILLS & INTERESTS

Technical Skills

Programming: Python (NumPy, Pandas, TensorFlow, scikit-learn, GPflow), R, C (basic)

Tools & Environments: Git/GitHub (version control), High-Performance Computing (HPC)
environments with SLURM job scheduling, CLI-based AI coding agents

Leadership, Teamwork & Personal Skills

Detail-oriented and accurate, with a strong sense of responsibility and inquisitiveness.

Proven ability to lead and collaborate effectively within teams.

Work well under pressure and deliver on tight deadlines while maintaining precision.

Languages

Italian (native), English (fluent, C1), Spanish (B2), Portuguese (basic)

Interests

Financial markets, Programming, Free-diving, Hiking, Chess